

A Packing of 12 Four-Spaces in $\mathbb{R}^{13}$

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Abstract

We give 12 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 3.020671. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.999702, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 12 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.020671 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 12$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.999702, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 12/11 = 3.020979$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N12.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 64 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
188791968913951437013036806888587509256175418009974394756178282324223131
695708840619777615422496195128798838329131574132511959765163938580618066
971419510184494825000212147145114367773490356728
```

denominator:

```
624999999998392433004721853440215010282970067801034977892460318201613488
442468091110591506627997422496496134952760081349969020509791361662536933
64001329908999435416028280133560733546595029525
```

B Coordinates

12 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N12.txt`.

```
-93505574485 -111262460054 -343695165426 1328376333456139656785 -126914270703 -129440085528 -19975566309293758352 131733199921 506483897361 -53787048838
-226350988885 -88009316830 70872923924 -1408116795006588424820 -120306630942 -209084306922 -3385322656055182791806 -228980789051 319906271780 402734445478
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